

Geometry of Splats

Definition (Chizat, Bach 2018 – informal)

Fix $\mu_0 \in \mathcal{P}(\Omega)$, $\Omega \subseteq \mathbb{R}^d$, and for $\mu \in \mathcal{P}(\Omega)$ set:

$$\mathcal{F}(\mu) = R \left(\int \Phi d\mu \right) + \int V d\mu$$

Assume (...) hold. Then, whenever the WGF $(\mu_t)_{t \geq 0} \rightarrow \mu_\infty$ in W_2 , also

$$\lim_{t \rightarrow \infty} \mathcal{F}(\mu_t) = \min_{\mu \in \mathcal{M}_+} \mathcal{F}(\mu)$$

Further: if $(\mu_{m,t})_{m \geq 1, t > 0}$ is a sequence of gradient flows $\mu_{m,0} \rightarrow \mu_0$ in W_2 , then limits exchangeable:

$$\lim_{m, t \rightarrow \infty} \mathcal{F}(\mu_{m,t}) = \min_{\mu \in \mathcal{M}_+} \mathcal{F}(\mu)$$

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Rotskoff et al. 2019: *conditional* global optimization of least squares risk over mean-field two layer net via W+FR dynamics.

Quantitative $O(t^{-1})$ decay of optimality gap, due to FR!